

WOLF CREEK PASS, CO, USA

WMO: 722101

Lat: 37.4628N

Lon: 106.8030W

Elev: 3594

StdP: 64.97

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 03039

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -20.2	(c) -17.9	(d) -37.7	(e) 0.2	(f) 0.7	(g) -32.9	(h) 0.3	(i) -0.5	(j) 16.0	(k) -12.6	(l) 14.3	(m) -9.5	(n) 5.6	(o) 220	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.7	(c) 17.8	(d) 6.3	(e) 17.1	(f) 6.1	(g) 15.9	(h) 5.9	(i) 9.2	(j) 14.0	(k) 8.4	(l) 13.2	(m) 7.7	(n) 12.4	(o) 4.2	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.5	10.1	10.1	6.4	9.3	9.3	5.9	9.1	9.1	37.8	13.8	35.2	13.0	33.2	13.0	12.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 15.1	(b) 13.0	(c) 11.8	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.6	-8.7	-7.9	-4.9	-1.7	2.2	9.4	11.8	10.0	6.9	1.4	-2.9	-8.3
	DBStd	8.20	4.74	4.62	4.42	3.87	4.12	3.11	2.23	2.46	3.22	4.12	4.62	4.64
	HDD10.0	3544	580	502	463	352	242	46	9	30	99	268	387	567
	HDD18.3	6456	838	735	722	602	500	267	203	257	343	526	637	825
	CDD10.0	131	0	0	0	0	1	29	64	31	6	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	5.6	5.6	6.8	6.7	6.7	6.1	5.3	4.1	4.0	5.2	5.4	5.9	5.9
Precipitation	PrecAvg	506	38	41	47	40	32	22	53	67	49	42	37	40
	PrecMax	634	85	103	104	90	72	52	91	139	86	111	67	86
	PrecMin	366	16	18	14	15	3	2	15	26	9	0	3	8
	PrecStd	62	17	16	19	22	18	12	20	25	19	24	15	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.0	3.0	7.5	9.7	14.0	18.7	21.1	18.8	16.2	12.4	8.9	3.0
		MCWB	-3.6	-3.7	-2.5	0.4	2.5	5.8	7.3	5.1	6.3	2.0	-0.5	-3.9
	2%	DB	0.3	1.2	4.8	7.1	12.1	17.4	18.0	17.3	14.1	9.8	7.0	1.0
		MCWB	-5.1	-4.2	-2.6	-1.2	2.0	5.2	6.9	6.2	4.3	0.4	-1.6	-4.8
	5%	DB	-1.0	-0.1	2.5	5.8	9.9	16.1	17.3	16.0	12.8	7.8	4.1	-0.7
		MCWB	-5.7	-4.3	-3.9	-1.5	0.9	4.5	6.6	6.1	4.3	0.5	-2.7	-5.5
	10%	DB	-2.4	-2.1	1.1	3.8	7.9	14.8	16.1	14.0	12.0	7.1	2.5	-2.3
		MCWB	-6.4	-5.4	-4.1	-2.5	0.3	4.0	6.5	5.9	4.1	0.1	-3.6	-6.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.4	-1.7	-0.4	1.3	4.3	8.1	10.5	10.0	7.8	5.9	0.9	-2.5
		MCDB	0.3	0.5	3.1	5.2	11.1	15.8	16.1	13.6	12.2	7.9	6.5	-0.8
	2%	WB	-3.5	-2.8	-1.8	0.2	3.2	7.0	9.7	9.0	6.9	4.1	-0.7	-3.5
		MCDB	-0.8	-1.2	2.3	4.4	9.6	14.7	15.0	12.7	11.4	6.7	4.5	-0.5
	5%	WB	-4.8	-3.8	-2.8	-0.7	1.9	6.1	8.8	8.2	6.1	2.3	-2.0	-4.6
		MCDB	-2.6	-1.1	1.0	3.6	7.5	13.5	14.1	12.0	10.7	5.7	2.6	-2.2
	10%	WB	-5.8	-4.9	-3.6	-1.4	1.0	5.3	8.1	7.5	5.4	1.2	-2.8	-5.5
		MCDB	-3.4	-2.2	0.0	3.0	6.5	12.1	13.2	11.5	9.7	4.5	1.1	-2.8
Mean Daily Temperature Range	5% DB	MDBR	6.5	6.5	7.3	8.0	7.5	8.6	8.7	8.0	7.7	7.0	6.5	6.3
		MCDBR	7.9	7.7	7.7	8.8	8.6	9.0	9.3	8.9	8.5	8.0	7.3	7.3
		MCWBR	5.4	5.3	4.7	5.0	4.5	4.9	4.5	4.7	4.4	5.0	4.2	4.9
	5% WB	MCDBR	6.8	6.4	6.6	7.7	8.0	8.7	8.8	7.9	7.8	6.6	6.8	6.6
		MCWBR	5.2	5.0	4.5	4.9	4.8	4.9	4.8	4.9	4.4	5.1	4.4	5.0
Clear-Sky Solar Irradiance	taub		0.188	0.192	0.211	0.233	0.245	0.243	0.281	0.266	0.238	0.218	0.203	0.191
	taud		2.608	2.616	2.616	2.634	2.668	2.694	2.599	2.633	2.773	2.786	2.701	2.626
	Ebn at Noon		1057	1083	1075	1054	1037	1032	991	1005	1031	1037	1030	1032
	Edh at Noon		68	78	87	91	90	88	96	90	74	65	62	62
All-Sky Solar Radiation	RadAvg		3.03	3.97	5.29	6.29	7.00	7.65	6.41	5.86	5.37	4.26	3.21	2.60
	RadStd		0.19	0.25	0.36	0.35	0.57	0.47	0.39	0.30	0.31	0.32	0.22	0.16

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (19 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page